HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

COURSE PROJECT REPORT

Course: Parallel Programming and Computing **Project title:** Parallel Video-Based People Counting with YOLO and OpenMPI on a Three-Machine Cluster **Group:** <ID nhóm> **Students:** Phạm Chí Bằng - 2035477; Nguyễn Thanh Lâm - 20235519; Nguyễn Phú An - 20235466; Lưu Hiếu An - 202400093 **Submission date:** 24 June 2026

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ABSTRACT

This project studies the parallelization of video-based people counting on a small physical cluster. A pretrained YOLO detector is used as the computationally intensive component, while the main contribution of the project is the design, implementation, and evaluation of a parallel inference pipeline using C++17 and OpenMPI. The system runs on three MacBook machines connected through a local area network. The benchmark mode uses CPU execution to align with the process-oriented requirements of the course, while a live-camera mode is kept as an application demonstration.

The video stream is decomposed along both time and image space. Frames provide independent temporal work units, and each frame may be divided into image regions to increase task granularity. These tasks are then mapped to MPI processes using static scheduling or a dynamic master-worker strategy. The master process gathers detector outputs, remaps bounding boxes to the original frame, removes duplicates across region boundaries, and produces frame-level people counts. This design exposes the main issues of parallel programming: decomposition, mapping, communication, load balancing, granularity, and speedup.

Experiments were conducted on MOT17-derived video data. The parallel implementation was first validated against a serial baseline, producing identical frame counts in the correctness test. The detector output was also compared against MOT17 ground truth counts to separate model accuracy from parallel correctness. A long-sequence benchmark identified a 600-frame workload whose wall-clock runtime was 123.667 seconds on twelve processes, satisfying the required two-to-three-minute input-size criterion. For a 1200-frame workload, the system achieved a wall-clock speedup of 1.939x at twelve processes. Additional experiments investigated granularity, scheduler behavior, and weighted process placement on a heterogeneous cluster.

Keywords: parallel computing, OpenMPI, object detection, YOLO, people counting, load balancing, speedup, distributed video processing.

TABLE OF CONTENTS

1. Introduction
2. Problem Statement and Background
3. Parallelization Methodology
4. Parallel Algorithm Design
5. Experimental Environment
6. Experimental Methodology
7. Results and Discussion
8. Compliance with Assignment Requirements
9. Limitations and Future Work
10. Conclusion
11. References

LIST OF FIGURES

1. Counting error over frames
2. Runtime as input size increases
3. Granularity overview
4. Per-process timing for the four-by-three configuration
5. Static and dynamic scheduling comparison
6. Speedup on the twelve-hundred-frame workload
7. Weighted mapping on the heterogeneous cluster

LIST OF TABLES

1. Physical cluster roles
2. Main experimental configuration
3. Parallel correctness
4. Detector accuracy against ground truth
5. Compact input-size trend
6. Long-sequence input-size trend
7. Granularity and load balance
8. Scheduler comparison
9. Speedup evaluation
10. Weighted mapping
11. Work distribution by host
12. Assignment requirement mapping

1. INTRODUCTION

People counting in video is a practical problem in classroom monitoring, public-space analysis, pedestrian flow estimation, and crowd management. Modern object detectors can identify people in individual frames, but applying a detector to a long video can be computationally expensive. Since many frames and image regions can be processed independently, the problem is a suitable case study for parallel programming on a distributed-memory cluster.

The aim of this project is not to train a new deep learning model. Instead, a pretrained YOLO detector is used to create a realistic and sufficiently heavy workload. The focus is the parallelization of the inference pipeline: how a video is decomposed, how tasks are assigned to processes, how intermediate detections are communicated, how duplicate boxes are removed, and how performance changes as the number of processes and task granularity vary.

The project also addresses an important distinction in applied parallel computing. A detector may produce inaccurate counts because of occlusion, small objects, lighting, or model limitations. This is a model accuracy issue. Separately, a parallel implementation may be incorrect if it changes the result compared with a serial implementation. This is a parallel correctness issue. The report therefore evaluates both aspects: serial-versus-parallel agreement and detector-versus-ground-truth accuracy.

The course assignment requires a physical MPI cluster of at least three machines, a meaningful parallel algorithm, a clear explanation of decomposition and communication, correctness verification, an input-size study, a granularity and load-balance study, and speedup measurements. This report follows that structure and presents experimental evidence for each requirement.

2. PROBLEM STATEMENT AND BACKGROUND

The problem is to estimate the number of people appearing in each frame of a video. The input is a video sequence and a pretrained object detector. The output is a sequence of predicted people counts, together with the detection boxes used to compute these counts. The computationally expensive operation is detector inference over many images or image regions.

YOLO is a one-stage object detection family. For each image, it predicts a set of bounding boxes, class labels, and confidence scores. In this project, only detections corresponding to the person class are retained. YOLO is not the subject of parallelization at the neural-network-kernel level; instead, the detector is treated as a local computational routine invoked by each MPI process. OpenMPI is responsible for distributing independent video tasks across processes.

MOT17 is used as the main dataset source because it is a common benchmark for pedestrian video analysis. It contains crowded scenes, occlusions, and people at multiple scales. These characteristics make it more challenging than a simple self-recorded video and more appropriate for evaluating a people-counting pipeline. A compact MOT17 subset is used for rapid correctness and granularity experiments, while longer sequences are used to find the required workload size and evaluate speedup.

The primary benchmark is CPU-based. This decision is intentional. The course requirement emphasizes MPI processes and processor assignment, so the main experiments must reflect process-level parallelism. Apple GPU and MPS execution are useful for a live demonstration, but they introduce GPU contention when multiple processes share the same device. Therefore, GPU execution is treated as an application demonstration rather than the official benchmark mode.

3. PARALLELIZATION METHODOLOGY

3.1 Level of Parallelism

The project uses task-level parallelism. Each unit of work corresponds to detector inference over one image region from one video frame. These units can be computed independently before their results are merged. Task-level parallelism is appropriate here because video processing naturally produces many independent pieces of work, and the runtime of each piece can vary depending on image content.

This differs from data-parallel training of neural networks. The project does not split a single neural-network operation across many processes. Instead, it distributes many independent detector calls across a cluster. MPI controls the process-level scheduling, while YOLO runs locally inside each assigned task.

3.2 Decomposition Technique

The decomposition is hybrid temporal-spatial decomposition. Temporal decomposition divides the video into frames. Spatial decomposition divides each frame into regions. A task is defined conceptually as one region from one frame. This hybrid design creates more work units than frame-only decomposition and therefore gives the scheduler more opportunities to keep all processes busy.

Frame-only decomposition is simpler, but it has weaknesses. If the video is short, the number of frames may not be enough to feed many processes. In addition, frames are not equally difficult: crowded frames and frames with severe occlusion may take longer to process. Spatial decomposition makes tasks finer, which can improve load balance, but it also increases communication and requires stronger post-processing to remove duplicated detections near region boundaries.

Thus, the decomposition is a trade-off. Coarse tasks reduce communication but may leave processes idle. Fine tasks improve scheduling flexibility but increase the amount of intermediate output and the cost of merging detections. The granularity experiment in this report measures this trade-off directly.

3.3 Mapping Technique

The two-dimensional structure of video frames and image regions is mapped into a one-dimensional list of independent tasks. This one-dimensional flattened mapping simplifies process assignment. The implementation supports two scheduling strategies.

In static scheduling, tasks are assigned in a deterministic block-cyclic manner before execution. This approach has low scheduling overhead and little communication during computation. However, it assumes that tasks have similar costs. If some processes receive harder regions or slower frames, the entire program must wait for them.

In dynamic scheduling, the master process maintains a task queue. Worker processes request work, return results, and receive more work until the queue is empty. The master may also perform computation so that it does not remain only a coordinator. Dynamic scheduling is more flexible for irregular workloads and heterogeneous machines, but it introduces additional communication and coordination overhead.

The project also includes a weighted process placement experiment. Since the machines are not identical, a stronger machine can be assigned more processes than a weaker one. This is not a replacement for the standard speedup experiment; it is an additional study showing how mapping can be adapted to a heterogeneous physical cluster.

3.4 Communication Strategy and Topology

The communication topology is a master-worker star topology. The master is responsible for task distribution, result collection, and final merging. Workers receive task descriptions, perform local detection, and send back detection results and timing metrics.

The communication strategy uses blocking communication for the main send and receive operations, together with probing at the master to detect available worker results. The gathered information includes both detection boxes and per-process timing data. The amount of data communicated is kept moderate because the video files and model files are synchronized to each machine before benchmarking. Therefore, the cluster mostly exchanges task metadata and detection results rather than raw video frames.

The master-worker design is suitable for this problem because duplicate removal and final counting require a global view of all detections from the same frame. If each process counted independently, a person crossing a region boundary could be counted more than once. Centralized merging at the master avoids this inconsistency.

3.5 Load Balancing Considerations

Load balancing is addressed at three levels. First, dynamic scheduling allows faster processes to receive additional work. Second, the project evaluates different region grids to control task granularity. Third, weighted mapping is tested to account for the different computational capacities of the machines.

The ideal system would keep all processes computing for most of the runtime. In practice, idle time can appear because tasks are uneven, the network is not free, the machines are heterogeneous, and the master must perform result merging. The report therefore measures computation time, communication time, and idle time for each process. This per-process view is more informative than a single total runtime.

4. PARALLEL ALGORITHM DESIGN

4.1 Overall Pipeline

The pipeline begins with video frame extraction. Each frame is optionally divided into several image regions. The scheduler assigns these regions to MPI processes. Each process runs YOLO locally on its assigned regions and returns the detected people boxes. The master then converts region-local detections into frame-level detections, removes duplicates, and computes the people count for each frame.

The serial baseline follows the same logical pipeline with a single process. This is important because the baseline and the parallel program use the same detector configuration and the same post-processing logic. A difference between the serial and parallel results would therefore indicate an error in task partitioning, communication, or merging rather than a difference in model behavior.

4.2 Static Scheduling

In static scheduling, all tasks are assigned before execution. Each process independently selects the tasks assigned to it and runs the detector. After finishing local work, the processes send their results to the master. The master combines all results, removes duplicated detections, and writes the final counts.

The main advantage of static scheduling is simplicity. It has lower coordination overhead because the master does not need to repeatedly dispatch work during execution. The main disadvantage is sensitivity to imbalance. If one process receives more expensive tasks, all other processes must wait at the end.

4.3 Dynamic Scheduling

In dynamic scheduling, the master initially sends work to available workers. Whenever a worker finishes, it sends results back and receives another task if any remain. The master can also process tasks locally, which makes the coordinator contribute computational work rather than only waiting for messages.

This strategy is more appropriate for irregular video inference. Some image regions contain many people, while others contain mostly background. Some machines are faster than others. Dynamic scheduling adapts to these variations by assigning more tasks to processes that finish earlier. The cost of this flexibility is more frequent communication and scheduling overhead.

4.4 Parallel Algorithm Pseudo-code

The following pseudo-code is written at the algorithmic level rather than as source code. It summarizes the behavior of the two scheduling strategies used in the project.

Algorithm 1. Static parallel people-counting pipeline

1. The master prepares the video workload by dividing the input sequence into frames and image regions.
2. All processes derive the same ordered task list from the workload description.
3. Each process selects the subset of tasks assigned to it by the static mapping rule.
4. Each process performs local detector inference on its assigned tasks and stores the resulting people detections.
5. All worker processes send their local detections and timing measurements to the master.
6. The master converts all detections to full-frame coordinates.
7. The master removes duplicated detections across neighboring regions.
8. The master computes the people count for each frame and writes the final experimental metrics.

Algorithm 2. Dynamic master-worker people-counting pipeline

1. The master prepares the same frame-and-region task list.
2. The master sends initial tasks to available worker processes.

3. A worker receives a task, performs local detector inference, and sends detections and timing measurements back to the master.
4. Whenever a worker returns a result, the master either assigns a new task or sends a termination message if no task remains.
5. If enabled, the master also processes tasks locally while waiting for worker results.
6. When all tasks are complete, the master gathers the final set of detections.
7. The master performs coordinate remapping, duplicate removal, and frame-level counting.
8. The master writes the output counts, detection records, and per-process timing metrics.

The two algorithms share the same detector and post-processing stages. Their main difference is task assignment. Static scheduling reduces dispatch overhead, while dynamic scheduling is more adaptive to uneven task costs and heterogeneous machines.

4.5 Bounding-Box Merging and Duplicate Removal

Spatial decomposition creates a post-processing challenge. A person near the boundary between two regions may be detected in both regions. Without global filtering, the same person could be counted multiple times. The master therefore performs a sequence of merging operations.

First, detections produced in region coordinates are transformed back into full-frame coordinates. Second, a region ownership rule keeps detections whose centers belong to the reliable core of the region. Third, global non-maximum suppression removes highly overlapping boxes across all regions of the same frame. Finally, additional duplicate handling reduces repeated detections for very large people close to the camera.

This post-processing stage is essential for correctness. The decomposition increases parallelism, but the final answer must still correspond to people in the original frame, not people separately counted in each region.

4.6 Timing Metrics

Each process records timing information in three categories: computation, communication, and idle or waiting time. Computation time represents local detector execution and local task processing. Communication time represents the exchange of task assignments, detections, and metrics. Idle time represents waiting caused by imbalance or synchronization.

The report distinguishes wall-clock runtime from computation-only runtime. Wall-clock runtime reflects the actual time experienced by the user and includes communication, coordination, and waiting. Computation-only runtime helps show how much time is spent on useful detector work. The gap between these two measurements indicates the cost of the distributed execution framework.

5. EXPERIMENTAL ENVIRONMENT

5.1 Physical Cluster

The experiments were run on three physical MacBook machines connected through the same local area network. Remote login was enabled on all machines, and OpenMPI was installed on each node. The cluster roles are shown in Table 1.

Role	Responsibility
Master	Coordinates execution, gathers results, stores outputs, and runs the live-camera display
Worker 1	Executes detector tasks and participates in benchmark runs
Worker 2	Executes detector tasks and participates in benchmark runs

The benchmark mode uses CPU execution. This makes the number of MPI processes meaningful for the course requirements. The live-camera mode may use Apple GPU acceleration for demonstration, but it is not mixed into the official CPU benchmark results.

5.2 Data and Model

The detector is a pretrained lightweight YOLO model. The main data source is MOT17. A compact MOT17 subset is used for quick correctness and granularity measurements. Longer MOT17-derived sequences are used for the input-size study and speedup measurement.

The ground truth annotations are converted into frame-level people counts. These counts are used only for the detector accuracy evaluation. They are not used for training. The project is therefore an inference and parallel-processing project, not a supervised training project.

5.3 Main Experimental Configuration

Table 2 summarizes the main configuration used for the reported CPU experiments.

Item	Setting
Detector	Pretrained YOLO
Benchmark device	CPU
Main scheduler	Dynamic master-worker scheduling
Main process count	Twelve processes
Main region grid	Four by three regions
Long-run region grid	Five by four regions
Dataset source	MOT17-derived sequences
Parallel framework	C++17 and OpenMPI

5.4 Code Volume

The implementation includes C++17 source code, Python helper utilities, plotting scripts, cluster scripts, and report-generation tools. The selected project files contain more than six thousand lines, exceeding the minimum code-size requirement for a four-member group. The report does not rely on line count as the main contribution; the main contribution is the parallel algorithm and its experimental evaluation.

6. EXPERIMENTAL METHODOLOGY

6.1 Parallel Correctness

The first experiment checks whether the parallel implementation preserves the serial result. The same video, detector, thresholds, and post-processing settings are used for both runs. The predicted count of each frame is compared between the serial baseline and the MPI implementation. A passing result means that the parallel decomposition, communication, and merging do not alter the answer produced by the pipeline.

6.2 Detector Accuracy Against Ground Truth

The second experiment compares the predicted frame counts against MOT17 ground truth counts. This measures detector accuracy rather than parallel correctness. The project reports mean absolute error, root mean squared error, percentage error, and exact-match ratio. These values are useful for understanding the application quality, but they should not be confused with correctness of the MPI implementation.

6.3 Input-Size Selection

The assignment requires an input size whose runtime is approximately two to three minutes when using a number of processes equal to the total number of physical cores selected for the cluster benchmark. The project measures several video lengths and reports runtime with and without communication. A 600-frame long-sequence workload is selected because its wall-clock runtime is 123.667 seconds with twelve processes.

6.4 Granularity and Load Balance

The granularity experiment varies the number of image regions per frame. The tested configurations range from a full-frame task to finer spatial divisions. For each configuration, the report records task count, maximum computation time, average computation time, total communication time, total idle time, and an idle-gap indicator. A stacked per-process chart is used to visualize computation, communication, and idle time.

The assignment specifies that if the idle time between any two processes differs by more than twenty-five percent, the system should be considered insufficiently balanced. The report explicitly applies this criterion and discusses why the measured cluster does not achieve ideal balance on the compact dataset.

6.5 Scheduler Comparison

Static scheduling and dynamic scheduling are compared under the same input setting. Static scheduling has less coordination overhead, while dynamic scheduling adapts better to uneven tasks and heterogeneous machines. The comparison is included to show that the choice of scheduler is not obvious: dynamic scheduling is more general, but it is not always faster for small inputs.

6.6 Speedup Evaluation

After selecting the required input size, the speedup experiment uses a workload twice as large. The process count is varied across one, two, four, eight, and twelve processes. Wall-clock speedup is computed from the ratio between the one-process runtime and the multi-process runtime. A separate computation-only trend is also shown to separate useful work from communication and coordination costs.

6.7 Weighted Mapping on a Heterogeneous Cluster

The cluster machines are not identical. A weighted mapping experiment assigns more processes to the stronger machine and fewer processes to the weaker one. This experiment is not the main course benchmark, but it provides additional insight into processor assignment on heterogeneous physical machines.

7. RESULTS AND DISCUSSION

7.1 Parallel Correctness

Table 3 shows the serial-versus-parallel correctness result. The MPI run produced the same frame counts as the serial baseline in the tested sequence.

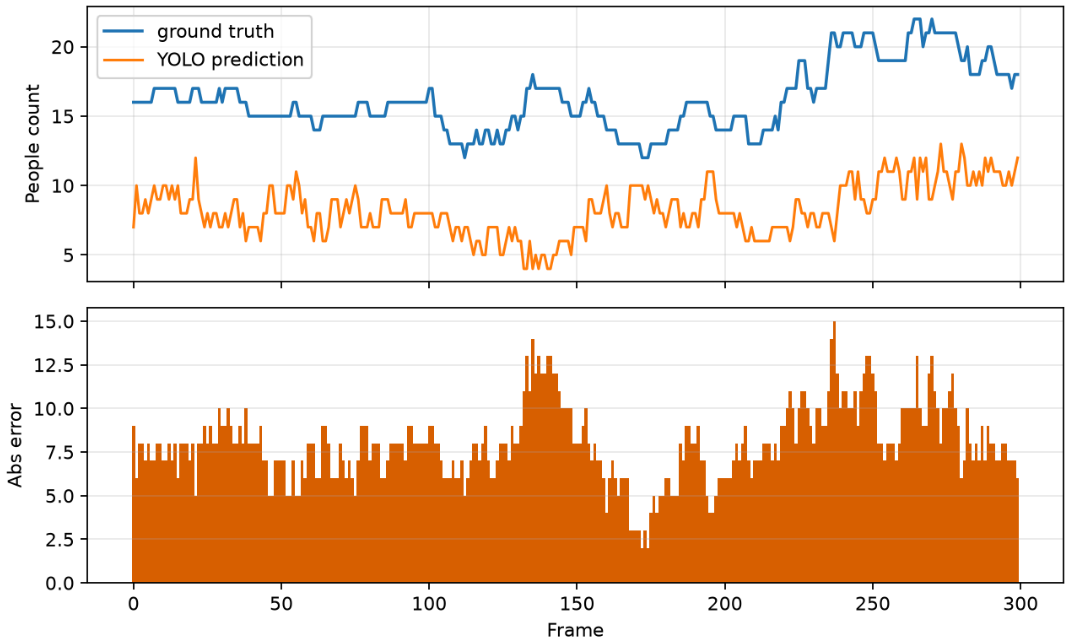
Result	Frames compared	Mismatched frames	Maximum count error	Mean count error
Passed	30	0	0	0.000

This result validates the parallel pipeline for the tested configuration. The frame-level agreement indicates that task splitting, result communication, coordinate remapping, and duplicate removal are consistent with the serial baseline.

7.2 Detector Accuracy

Table 4 compares predicted counts against MOT17 ground truth counts. The detector undercounts in crowded scenes, which is expected because the model is pretrained and not specialized for the MOT17 crowd setting.

Frames	Mean absolute error	Root mean squared error	Percentage error	Exact-match ratio	Ground-truth average	Predicted average
300	7.983	8.269	0.490	0.000	16.207	8.223



Counting error over frames

The accuracy result should be interpreted separately from the parallel correctness result. The model may undercount people because of occlusion, small pedestrians, and crowded scenes, but the MPI implementation can still be correct if it reproduces the serial pipeline output.

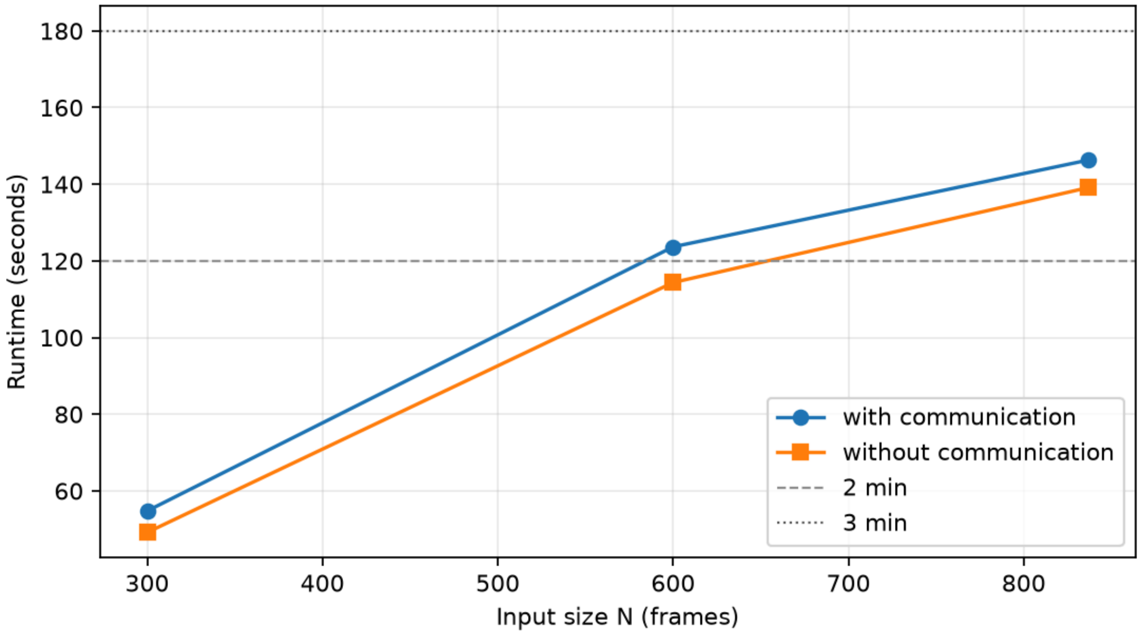
7.3 Input-Size Study

The compact MOT17 subset is useful for quick tests but is too short to reach the required two-to-three-minute runtime. Table 5 reports the compact-sequence trend.

Frames	Runtime with communication (s)	Runtime without communication (s)	Processes	Region grid
30	9.165	2.999	12	Four by three
60	12.570	6.801	12	Four by three
100	17.776	11.596	12	Four by three
150	23.585	17.103	12	Four by three

The long-sequence experiment uses a finer region grid and longer video segments. Table 6 shows the result.

Frames	Runtime with communication (s)	Runtime without communication (s)	Processes	Region grid
300	54.764	49.156	12	Five by four
600	123.667	114.352	12	Five by four
837	146.316	139.136	12	Five by four



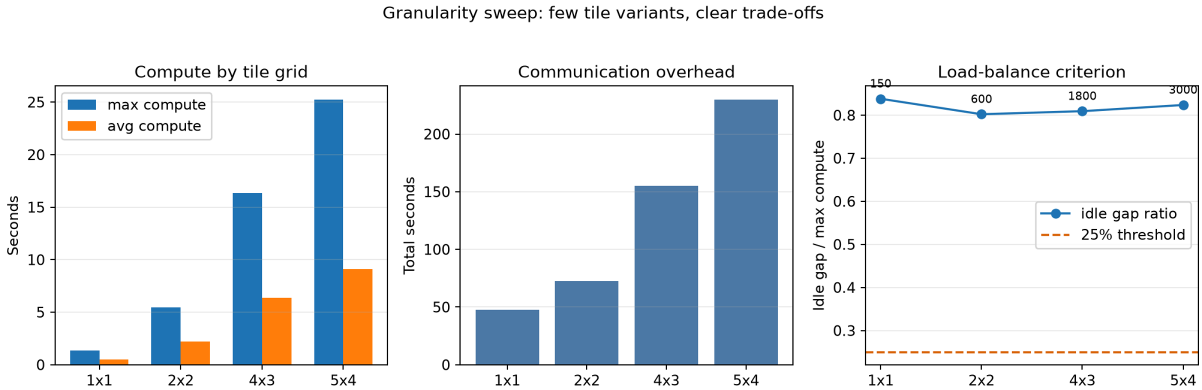
Runtime as input size increases

The selected input size is six hundred frames. Its wall-clock runtime is 123.667 seconds, which lies within the required two-to-three-minute interval. The difference between the two runtime curves represents the cost of communication, coordination, and waiting.

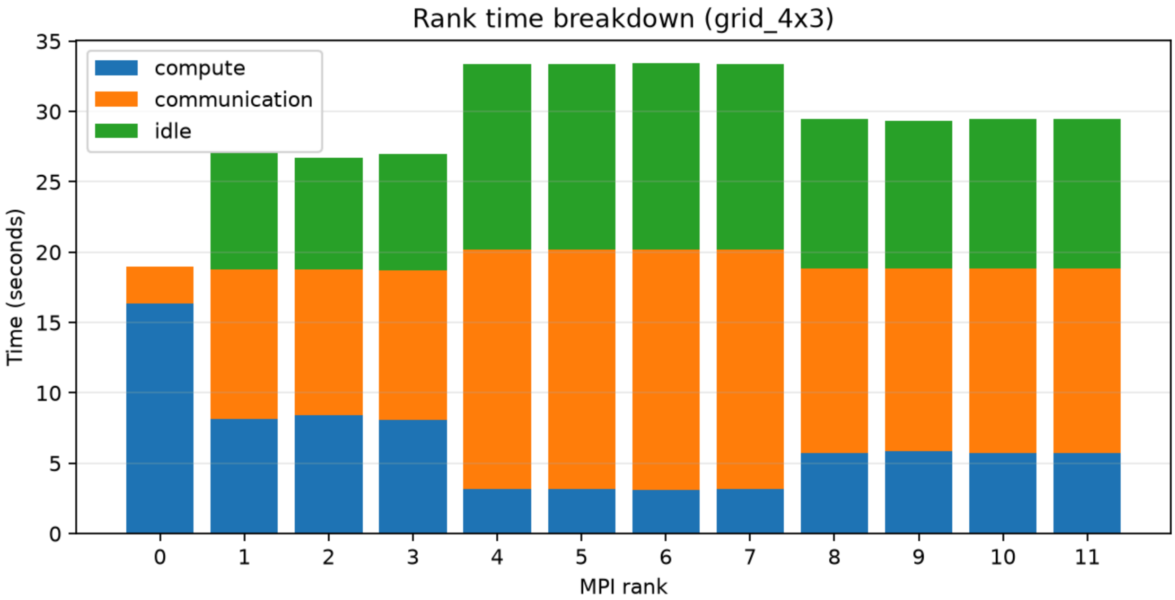
7.4 Granularity and Load Balance

Table 7 summarizes the granularity experiment. The compact dataset does not achieve the desired idle-balance criterion, but the result is useful because it reveals how task size affects computation, communication, and waiting.

Region grid	Processes	Tasks	Maximum computation (s)	Average computation (s)	Total communication (s)	Total idle (s)	Idle-gap indicator	Balance result
One by one	12	150	1.367	0.483	47.417	10.615	0.838	Not balanced
Two by two	12	600	5.447	2.205	72.434	38.902	0.802	Not balanced
Four by three	12	1800	16.329	6.375	154.944	119.449	0.809	Not balanced
Five by four	12	3000	25.263	9.133	230.059	193.569	0.824	Not balanced



Granularity overview



Per-process timing for the four-by-three configuration

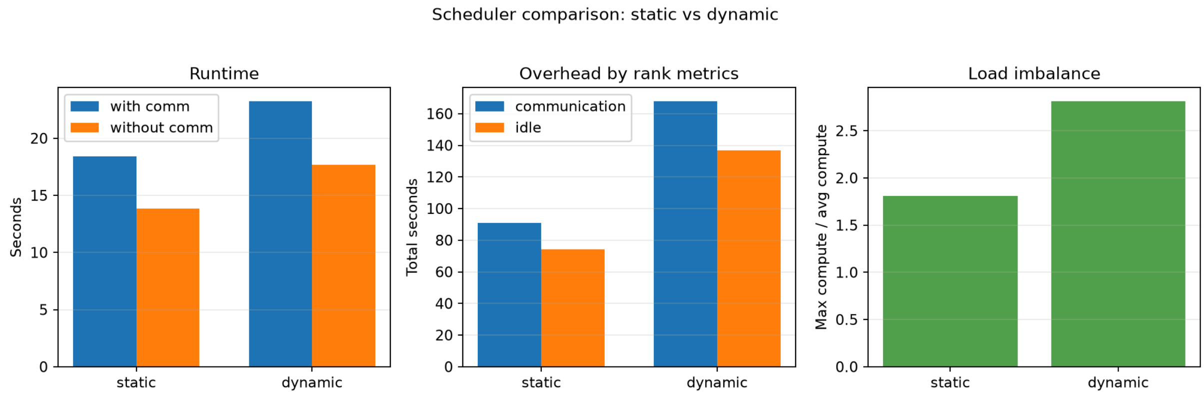
The full-frame setting is too coarse to expose much spatial parallelism. Finer regions create more tasks, which gives the scheduler more opportunities to distribute work. However, finer regions also create more intermediate detections and more communication. The four-by-three setting is a reasonable demonstration configuration, while the five-by-four setting is more suitable for the longer input-size and speedup experiments.

The balance result is not ideal. Rather than hiding this, the experiment makes the limitation visible. The likely causes are heterogeneous machines, short compact input, detector-worker overhead, and master-side merging. This is a realistic outcome in distributed-memory video processing and motivates the scheduler and weighted-mapping experiments.

7.5 Static and Dynamic Scheduling

Table 8 compares static and dynamic scheduling on the same compact input.

Scheduler	Processes	Frames	Region grid	Runtime with communication (s)	Runtime without communication (s)	Load imbalance	Idle-gap indicator	Balance result
Static	12	150	Four by three	18.437	13.823	1.807	0.754	Not balanced
Dynamic	12	150	Four by three	23.262	17.699	2.815	0.826	Not balanced



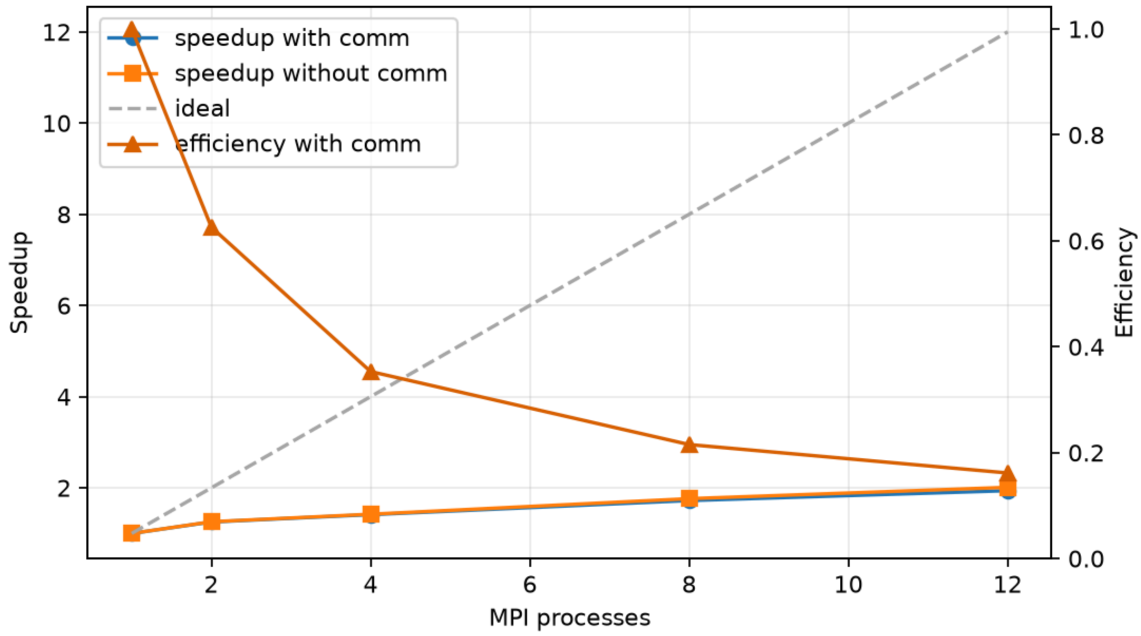
Static and dynamic scheduling comparison

Dynamic scheduling is not faster in this compact experiment because its coordination overhead is significant relative to the amount of work. This is an important observation: a more flexible scheduler is not automatically faster. Dynamic scheduling remains valuable because it generalizes better to irregular workloads and heterogeneous machines, but the input must be large enough for its benefits to overcome dispatch overhead.

7.6 Speedup on a Workload Twice as Large

After selecting the six-hundred-frame workload, the speedup experiment uses a twelve-hundred-frame input. Table 9 shows the runtime and speedup trend.

Processes	Runtime with communication (s)	Runtime without communication (s)	Wall-clock speedup	Efficiency
1	345.677	342.467	1.000	1.000
2	276.155	272.546	1.252	0.626
4	244.747	240.905	1.412	0.353
8	200.770	194.282	1.722	0.215
12	178.306	170.255	1.939	0.162



Speedup on the twelve-hundred-frame workload

Speedup improves as the number of processes increases, but it is not linear. At twelve processes, the wall-clock speedup is 1.939x. The computation-only speedup is approximately 2.011x. The gap between these values is caused by communication, scheduling, result collection, duplicate removal, and contention for CPU resources on each machine.

This result is reasonable for a real cluster running a mixed workload. Pure numerical kernels often scale better because they have less irregular post-processing and smaller coordination overhead. In contrast, this project combines detector inference, video processing, distributed communication, and global merging.

7.7 Weighted Mapping on Heterogeneous Machines

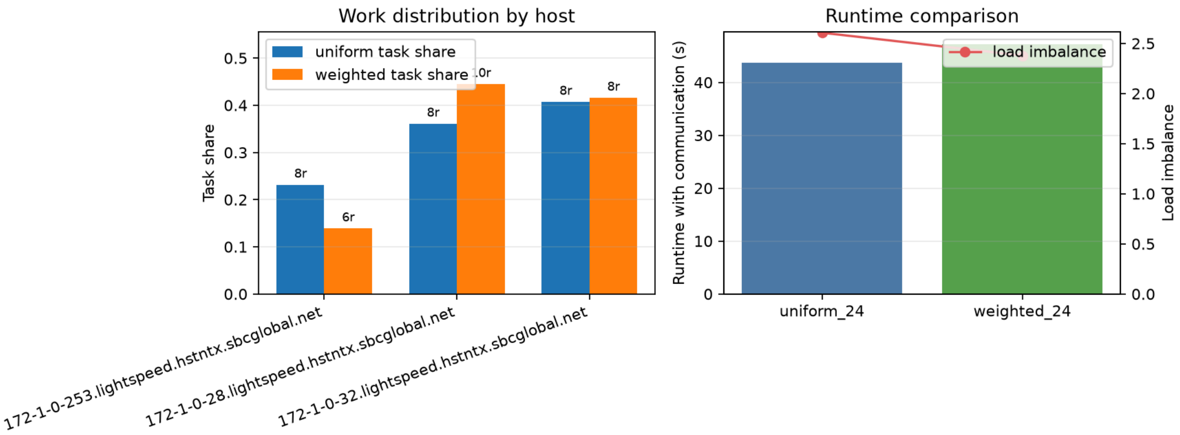
Table 10 shows the weighted mapping experiment using twenty-four processes. The weighted placement gives more processes to the stronger machine and fewer to the weaker one.

Mapping	Processes	Frames	Region grid	Runtime with communication (s)	Runtime without communication (s)	Load imbalance
Uniform placement	24	150	Five by four	43.659	32.268	2.607
Weighted placement	24	150	Five by four	47.238	30.864	2.379

Table 11 shows how work was distributed across hosts.

Mapping	Host category	Assigned processes	Processed tasks	Task share
Uniform placement	Weaker worker	8	695	0.232
Uniform placement	Stronger worker	8	1083	0.361

Mapping	Host category	Assigned processes	Processed tasks	Task share
Uniform placement	Master	8	1222	0.407
Weighted placement	Weaker worker	6	417	0.139
Weighted placement	Stronger worker	10	1335	0.445
Weighted placement	Master	8	1248	0.416



Weighted mapping on the heterogeneous cluster

Weighted placement improves computation-only time and reduces measured load imbalance. However, wall-clock runtime increases because communication and coordination costs become larger. This shows that assigning more processes to stronger hardware can help, but only if the communication overhead remains controlled. The result also supports the broader conclusion that process count alone is not a sufficient measure of parallel performance.

7.8 Summary of Experimental Findings

The experiments support four main conclusions. First, the MPI implementation preserves the serial pipeline output in the correctness test. Second, the detector accuracy on MOT17 is limited by the pretrained model and the difficulty of crowded scenes, not by MPI parallelization. Third, the selected six-hundred-frame input satisfies the required runtime interval, and the twelve-hundred-frame workload shows measurable speedup. Fourth, granularity and mapping strongly affect performance, but neither finer tasks nor more processes automatically guarantee better runtime.

The results also show that communication and synchronization are central issues in this project. The algorithm contains a sequential merging component at the master, and distributed execution introduces coordination overhead. These factors limit speedup, but they are precisely the factors that make the project relevant to parallel computing rather than merely object detection.

7.9 Live-Camera Demonstration

The project includes a live-camera mode in which the master captures video from its camera, distributes computation to the cluster, and displays the detected people count in real time. This mode demonstrates the application value of the system. However, the formal benchmark results in this report use offline MOT17-derived videos and CPU execution to keep the performance evaluation reproducible and aligned with MPI process-level requirements.

8. COMPLIANCE WITH ASSIGNMENT REQUIREMENTS

Table 12 maps the assignment requirements to the project outcomes.

Requirement	Project response
At least three physical machines	The system runs on three MacBook machines in the same local network
No cloud servers	All experiments were conducted on physical machines owned by the group
Parallelization level	Task-level parallelism
Decomposition method	Hybrid temporal-spatial decomposition
Mapping technique	One-dimensional flattened task mapping with static, dynamic, and weighted variants
Communication strategy	Master-worker star topology with MPI communication
Load balancing	Dynamic scheduling, granularity study, and weighted process placement
Parallel algorithm description	Static and dynamic algorithms are described in the methodology section
Correctness	Serial and MPI outputs match in the correctness experiment
Input-size selection	A six-hundred-frame workload runs in 123.667 seconds
Granularity evaluation	Multiple region grids are evaluated with per-process timing charts
Speedup evaluation	One, two, four, eight, and twelve processes are tested on a workload twice as large
Report length	The exported PDF is checked to remain between ten and twenty pages
Code-size requirement	The project implementation exceeds the required line count for four members

9. LIMITATIONS AND FUTURE WORK

The first limitation is detector accuracy. The pretrained model undercounts crowded MOT17 scenes. Fine-tuning on pedestrian and crowd data or using a larger detector could improve application accuracy, although it would also increase runtime.

The second limitation is communication overhead. Dynamic scheduling is more flexible, but frequent task dispatch and result collection can become expensive. A future version could batch multiple tasks per dispatch, use more non-blocking communication, or overlap communication with local post-processing more aggressively.

The third limitation is duplicate handling. Spatial decomposition requires careful merging near region boundaries. The current global suppression and ownership rules reduce duplicates, but difficult camera-close cases can still create false positives or false merges. A temporal tracker such as ByteTrack or DeepSORT could stabilize counts across frames.

The fourth limitation is hardware heterogeneity. The cluster contains machines with different performance levels. Weighted placement helps compute-only time, but it can increase wall-clock time if communication grows. A more advanced scheduler could estimate machine speed online and adapt task assignment continuously rather than relying only on fixed process placement.

Finally, the experiments were conducted on a small three-machine cluster. This is appropriate for the course requirement, but larger clusters would require additional analysis of network bottlenecks, master scalability, and distributed result aggregation.

10. CONCLUSION

This project implemented and evaluated a parallel people-counting pipeline using YOLO and C++17/OpenMPI on a three-machine physical cluster. The algorithm uses task-level parallelism with hybrid temporal-spatial decomposition. Static scheduling, dynamic scheduling, and weighted process placement were implemented and experimentally studied.

The parallel implementation passed the serial-versus-MPI correctness test. A 600-frame workload was selected because its twelve-process runtime was 123.667 seconds, satisfying the required two-to-three-minute range. On a 1200-frame workload, the system achieved a wall-clock speedup of 1.939x at twelve processes. The granularity and scheduler experiments showed that task size, communication overhead, and hardware heterogeneity strongly influence performance.

Although the speedup is not linear, the project demonstrates the central concepts of parallel programming: decomposition, mapping, communication, load balancing, correctness, granularity, and performance evaluation. The live-camera mode further shows that the system can be used as an interactive distributed vision application, while the formal CPU benchmarks provide reproducible evidence for the course report.

11. REFERENCES

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APPENDIX A. PRESENTATION NOTES

If asked about the level of parallelism, the answer is task-level parallelism. If asked about decomposition, the answer is hybrid temporal-spatial decomposition. If asked about mapping, the answer is a one-dimensional flattened mapping with static, dynamic, and weighted variants. If asked about communication topology, the answer is master-worker star topology. If asked why detector accuracy is imperfect, the answer is that YOLO model accuracy is different from parallel correctness; serial and parallel outputs match, while model-versus-ground-truth accuracy depends on the pretrained detector and dataset difficulty.